

Engineering Plasmonic Surface Lattice Resonances via a Nanoparticle Filling Factor

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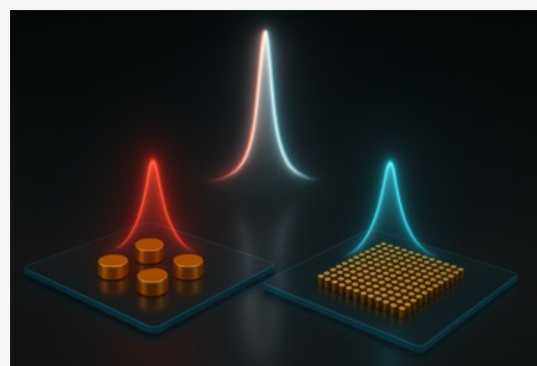
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ABSTRACT: The theory of dielectric photonic crystals (PCs) has been extensively developed, with band diagrams providing key insights into their eigenmodes. Under certain conditions, the band structures are scale-invariant; i.e., the relative positions of the bands do not change if the lattice geometry is uniformly scaled. This implies that the PC can operate similarly across different size regimes. In contrast, arrays of metallic nanoparticles exhibit dispersive behavior, but for surface lattice resonance (SLR) modes in the coherent coupling regime, their behavior appears to be similar to that of dielectric PCs. In this work, we combine experimental studies with finite-difference time-domain (FDTD) simulations to investigate the influence of the filling factor (ratio of the nanoparticle surface area to that of the unit cell) in determining the SLR wavelength. We demonstrate that two distinct 2D nanoparticle periodic arrays with identical filling factors and lattice periods along the propagation direction of the Rayleigh anomaly support SLRs at the same wavelength in the coherent coupling regime. Insight into the filling factor in SLRs provides a simple geometrical rule to control plasmonic resonances, enabling the rational design of substrates for sensing, photocatalysis, and surface-enhanced spectroscopies.



The theory of dielectric photonic crystals (PCs) has been extensively developed and applied.¹ These dielectric PCs are typically characterized by their photonic band diagrams, which allow the extraction of a wealth of information about the structure's eigenmodes, including their spectral positions, effective indices, group velocities, etc.² These band diagrams are governed by the filling factor, the index contrast, and the structure's geometry. They are generally calculated for a given filling factor and are represented in the first irreducible Brillouin zone as a function of the frequency normalized with respect to the lattice period. In fact, the Helmholtz eigenvalue equation exhibits an important property: it remains invariant under scaling if the refractive index is independent of the frequency. As a result, for a given filling factor that directly impacts the position and width of the band gap, the shape of the band structure for infrared waves is the same as for the visible spectrum.³

Conversely, metallic nanoparticle gratings exhibit more pronounced dispersive behavior in the optical range than in the microwave range. Such nanoparticles absorb and scatter light through localized surface plasmon resonances (LSPRs)—coherent oscillations of conduction electrons driven by the incident electric field, which enhance the optical near field.^{4–9} The LSPR frequency depends strongly on nanoparticle size, shape, and the permittivity of both the metal and surrounding medium.^{10–13} Consequently, it is not possible to plot their

corresponding band diagrams as a function of the reduced frequency. Similarly, the filling factor is not very relevant, as plasmonic modes and their associated Joule losses depend partially on the size of the nanoparticles. For gratings with large interparticle distances (on the scale of a wavelength), long-range radiative coupling creates hybrid photonic modes with delocalized nature.^{14–18} These delocalized photonic modes result in narrow and Fano-like resonances in the far-field response,¹⁹ known as plasmonic surface lattice resonances (SLRs). Over the past decade, mastering plasmon coupling in surface lattice resonances (SLRs) has emerged as a central focus, enabling breakthroughs in biosensing, lasing, and enhanced spectroscopy.^{20–22}

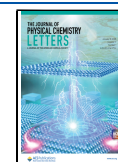
Interestingly, the theoretical background associated with dielectric PCs can be used to study the SLR modes. Indeed, when the SLR mode is strongly delocalized (coherent coupling with lattice diffraction modes, the so-called Rayleigh anomalies), it becomes highly photonic, and its spatial

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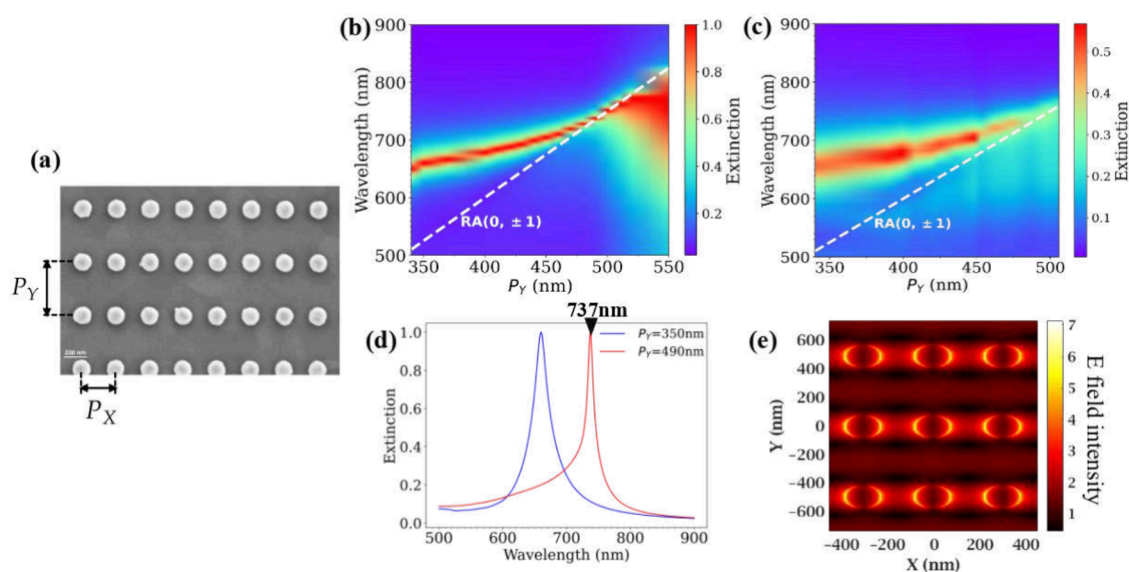


Figure 1. (a) SEM view of one of the studied structures. All of the considered structures are composed of gold nanodisks with height H of 50 nm and diameters D of 120 nm or 160 nm. The nanodisks are surrounded by air and deposited on a glass substrate covered with an 80 nm thick ITO layer. (b) and (c) Dispersion diagrams as a function of lattice constant in the y direction and wavelength. (b) Obtained by FDTD calculation and (c) by extinction measurement. Both were performed for lattices consisting of nanodisks with a diameter of 160 nm and an x -direction pitch of 300 nm. The structures are illuminated by a Gaussian beam with linear polarization in the x direction at normal incidence from the substrate, and measurements are taken in air. (d) Calculated extinction spectra of two arrays with the same parameters as in (b), with $P_y = 350$ nm and 490 nm. (e) Mapping, performed just above the nanoparticles, of the energy distribution calculated at the wavelength of the maximum extinction spectrum corresponding to the array shown in red in (d).

extension is almost entirely within the surrounding dielectric medium.^{9,23} Consequently, its surface propagation is mainly governed by the medium's refractive index. Under the right conditions, gratings with the same filling factor are expected to show SLR eigenmodes at the same wavelength. In this work, we aim to conduct an experimental study supported by FDTD simulations to demonstrate that in the coherent coupling regime with the Rayleigh anomaly the SLR spectral position is primarily governed by the filling factor. We show, in particular, that two distinct 2D nanoparticle periodic arrays with the same filling factor and the same lattice period along the propagation direction of the anomaly support SLRs at the same wavelength in the coherent coupling region. Beyond the well-known effect of particle size, we identify the filling factor as an additional geometric parameter that governs SLR wavelengths across distinct nanoparticle shapes and 2D arrays. This finding establishes a simple geometric principle to guide the rational design of plasmonic nanostructures for chemical sensing or photocatalysis.^{24,25}

To investigate the influence of the filling factor, we consider rectangular arrays of gold nanodisks of diameter D and fixed height $H = 50$ nm, fabricated using electron beam lithography (EBL).^{26,27} The lattice periods in the x and y directions are denoted by P_x and P_y , respectively (see Figure 1(a) displaying an SEM image of one of the studied structures). Gold nanodisks are deposited on a glass substrate ($n = 1.5$) coated with an 80 nm thick layer of indium tin oxide (ITO) which acts as a superstrate and is surrounded by air. Light illuminates the particle array from below, transmitting through the ITO to the particles by a plane wave polarized along the x -axis. This polarization direction enables the excitation of grazing orders and Rayleigh anomalies propagating in the y direction. The transmitted signal is collected in air and processed to determine the extinction spectra. For an incident wave

polarized along the x -axis, the position of the Rayleigh anomaly $(0, \pm 1)$ due to the substrate, which couples with the LSPR to produce an SLR, is determined by the lattice period P_y , the incidence angle, and the substrate index. At normal incidence, it is given by

$$\lambda_{\text{RA}}(0, \pm 1) = n_{\text{sub}} \cdot P_y$$

In our 2D periodic nanoparticle arrays, the $(0, \pm 1)$ diffracted order is initially evanescent and couples with the LSPR to produce a surface lattice resonance. As the wavelength approaches the Rayleigh anomaly, the diffracted order becomes grazing along the plane, corresponding to the region of maximal coupling. Beyond this point, the order becomes propagative out of plane, and the coupling ends. To identify the zone of coherent coupling of LSPR with the Rayleigh anomaly $(0, \pm 1)$, we first calculated and measured the extinction spectra of the nanodisk arrays by varying the period P_y from 300 nm to 600 nm. The particle diameter and x -axis period were kept constant at $D = 160$ nm and $P_x = 300$ nm, respectively. These gratings will be referred to as “type I” gratings hereafter. In Figures 1(b) and 1(c), we plotted the calculated and measured extinction diagrams as a function of period P_y . The optical response of the samples was analyzed in air, using far-field visible-NIR extinction microspectroscopy in the wavelength range 400–1000 nm. Illumination was provided by a halogen lamp from the glass side at a normal incidence. The spectrometer (LOT ORIEL model MS 260i) was coupled to an optical microscope (OLYMPUS BX 51) equipped with a 10× objective lens (numerical aperture, N.A., 0.25). It is worth noting the very good agreement between the experimental and the FDTD-calculated dispersion diagrams.

These diagrams show an SLR plasmon resonance band red-shifting as the period P_y increases. Furthermore, as can be seen from the diagrams, the width of the band narrows as the

spectral position of the Rayleigh anomaly due to the substrate approaches the SLR position. This indicates that the area of coherent coupling lies within the $470 \text{ nm} \leq P_y \leq 490 \text{ nm}$ range. Beyond this range, the anomaly is less energetic than the SLR, which reverts to a simple LSPR with a broader, asymmetric profile characterized by a pronounced shoulder at shorter wavelengths (Figure 1(b)).

As shown in Figure 1(d), the extinction spectrum corresponding to $P_y = 490 \text{ nm}$, which has a spectral position very close to the Rayleigh anomaly position, is much sharper than that relating to $P_y = 350 \text{ nm}$, whose spectral position is far from that of the Rayleigh anomaly. The sharpness of the resonance at $P_y = 490 \text{ nm}$ reflects the low energy loss of the SLR during propagation, which is due to the low penetration of the mode into the gold NPs.

In the zone of coherent coupling, the SLR mode is mainly photonic and delocalized on the grating surface, as confirmed by electrical intensity mapping, calculated for $P_y = 490 \text{ nm}$ and plotted at $\lambda = \lambda_{\text{SLR}}$ (Figure 1(e)) in the (x, y) plane just above the metal particles.

Owing to its delocalized nature, the SLR mode is an eigenmode of the structure, identical to the Bloch mode of the photonic structure, excited at $\lambda = \lambda_{\text{SLR}}$ by an incident plane wave. In order to determine the conditions under which the theoretical background of dielectric photonic crystals can be applied to metal gratings supporting delocalized SLRs, photonic band calculations were performed on two plasmonic gratings with the same filling factor. One grating is of type I, while the other has geometrical parameters of $D = 120 \text{ nm}$ and $P_x = 170 \text{ nm}$ and will be referred to as a type II grating henceforth. We have also performed a band calculation on a third type of grating (type III) with parameters of $D = 160 \text{ nm}$ and $P_x = 210 \text{ nm}$, which have different filling factors.

It should be noted that the filling factor in the case of rectangular arrays is given by

$$f = \left(\frac{\pi R^2}{P_x} \right) \cdot \frac{1}{P_y}$$

The geometric parameters of the arrays under consideration are summarized in Table 1. Although the filling factors (in $1/P_y$

Table 1. Parameters of the Studied Arrays

Array	$D \text{ (nm)}$	$P_x \text{ (nm)}$	$P_y \text{ (nm)}$	f
I-type	160	300	variable	$67.02 \times \frac{1}{P_y}$
II-type	120	170	variable	$66.52 \times \frac{1}{P_y}$
III-type	160	210	variable	$95.74 \times \frac{1}{P_y}$

units) for I- and II-type arrays are not exactly equal (see Table 1), we can consider these values to be the same when taking into account the FDTD discretization step and the size dispersion of the manufactured arrays resulting from slight imperfections introduced during the EBL fabrication process.

In Figure 2(a), we plotted the normalized spectral energy density calculated at the Γ point for these three gratings. The period P_y is kept constant for these gratings and is set at 490 nm , which corresponds to the zone of coherent coupling of the I-type grating that has already been identified. This choice also

guarantees that the SLR resulting from the Rayleigh's $(0, \pm 1)$ anomaly coupling is excited on all three arrays. As Figure 2(a) shows, I- and II-type gratings have a Bloch eigenmode with the same profile and located at the same wavelength ($\lambda = 737 \text{ nm}$). In contrast, the third grating exhibits a very broad eigenmode, with its spectral position red-shifted to $\lambda = 766 \text{ nm}$. This lies far from Rayleigh's $(0, \pm 1)$ anomaly and outside the coherent coupling zone, despite two of its three parameters (D and P_y) being identical to those of the type-I grating.

The FDTD extinction spectra of these three gratings, shown in Figure 2(b), exhibit a similar behavior. The extinction spectra of type-I and -II gratings are narrow and almost merged, whereas the third grating is broad. Their spectral positions are identical to those predicted by the band calculation. Therefore, the extinction maxima that determine the positions of SLRs correspond well to Bloch modes excited by a linearly polarized plane wave at normal incidence. As expected for a photonic band calculation on dielectric gratings, these results show that SLR resonances in the coherent coupling zone are dominated by their photonic character. In other words, their electromagnetic energy is strongly delocalized from the metal, thereby reducing Joule losses. These results clearly demonstrate that the exact position of these delocalized SLR modes in the coherent coupling zone can also be governed by the filling factor. Furthermore, the sharp profile of the energy density spectra obtained by band calculation is a strong indicator of the photonic and delocalized nature of the SLR modes.

To further confirm the photonic nature of SLR in the coherently coupled zone, we measured the experimental extinction spectra of gratings of types I and II (the same filling factor, given P_y) and superimposed them on each subfigure of Figure 3 (Figure 3(a–d)). FDTD simulations are shown in Figure 3(a'–d').

First, we observe a good agreement between the experimental and theoretical spectra. When we move away from the coherent coupling zone, corresponding to a P_y lower than 470 nm (see Figures 3(a–b), as well as 3(a'–b')), the extinction spectra of the two gratings exhibit broad profiles and do not have the same resonance wavelength. This behavior reflects the strong plasmonic character of the SLR resonances supported by the two types of grating and the difference in the specific surface areas of the nanodiscs that constitute these gratings. In the region of coherent coupling, corresponding to $P_y = 470 \text{ nm}$ and 490 nm (Figures 3(c–d) and 3(c'–d')), the measured and calculated spectra of the two gratings overlap almost perfectly. This reflects the preponderance of the photonic character of the SLR resonance in this region of coherent coupling, which is mainly dependent on the nondispersive dielectric medium surrounding the metal nanodiscs. It is worth noting that the broad feature observed in the experimental spectra around $620\text{--}630 \text{ nm}$ corresponds to the dipolar plasmonic resonance of the isolated nanodisk.

Figure 4 compares the evolution of extinction spectra maxima (SLR wavelengths) with P_y for the I- and II-type gratings. The red circles correspond to the maxima of type-I gratings, and the blue circles correspond to the maxima of type-II gratings. The vertical bars represent the full width at half-maximum (fwhm) of the different resonances. Each of these bars represents the sum of the resonance widths on both sides: toward longer wavelengths (right) and shorter wavelengths (left) of the peak. The green vertical dashed lines

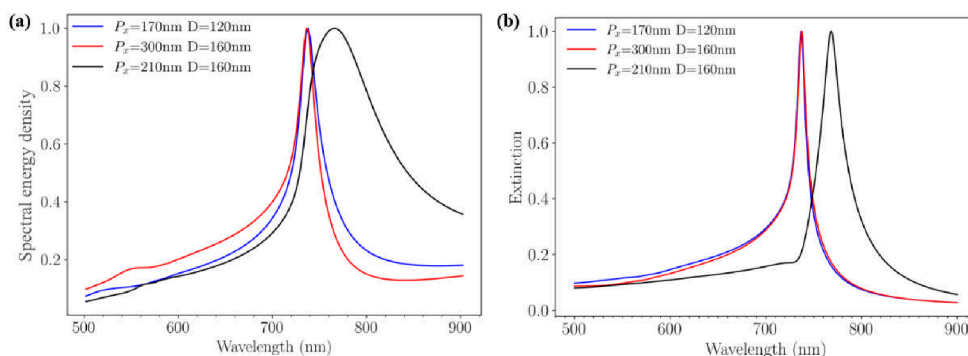


Figure 2. (a) Energy spectral densities calculated at the Γ point and (b) FDTD extinction spectra of three arrays of gold nanodisks, all with the same P_y set at 490 nm. The blue, red and black curves correspond to the following parameter pairs, respectively: ($P_x = 170$ nm and $D = 120$ nm), ($P_x = 300$ nm and $D = 160$ nm), and ($P_x = 210$ nm and $D = 160$ nm).

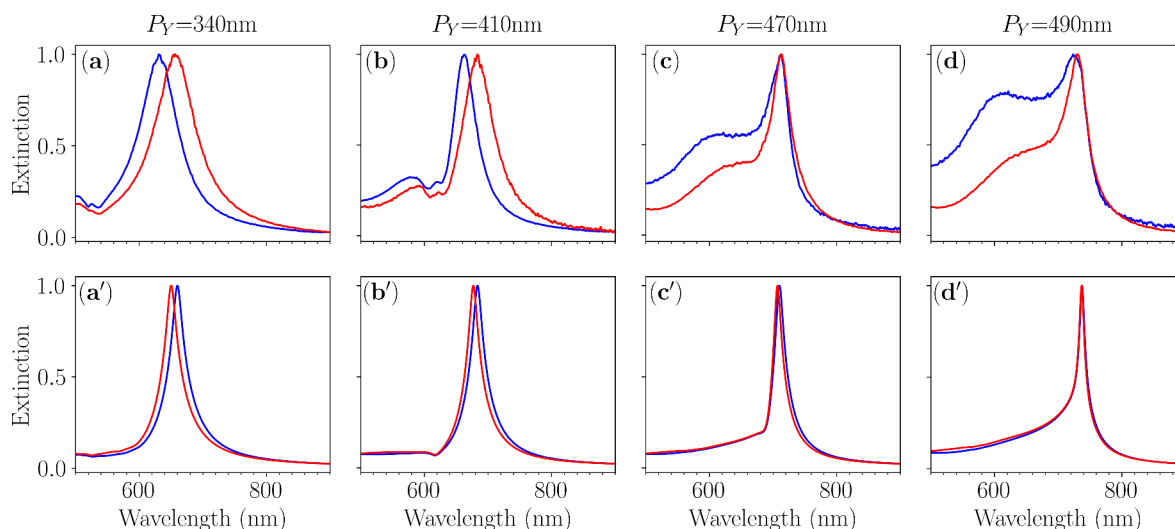


Figure 3. (a–d) Experimental and (a'–d') FDTD calculated extinction spectra of nanodisk arrays of diameter $D = 120$ nm and $P_x = 170$ nm (in blue line) and nanodisk arrays of diameter $D = 160$ nm and $P_x = 300$ nm (in red line) and for values of P_y of 340 nm, 410 nm, 470 nm, and 490 nm. The figure highlights the evolution of the extinction spectra with increasing P_y , showing the progressive red-shift of the surface lattice resonance modes. The experimental and simulated spectra exhibit consistent trends, confirming the validity of the FDTD model.

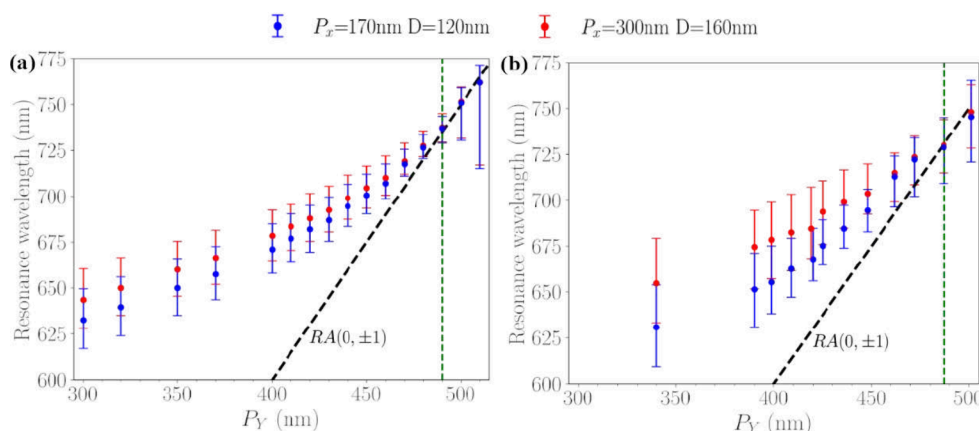


Figure 4. Evolution of the spectral positions and full width at half-maximum (fwhm) of SLRs as a function of P_y , deduced from (a) FDTD calculated and (b) measured extinction spectra for arrays of type I (red) and type II (blue).

indicate the upper limit ($P_y = 490$ nm) of the coherent coupling region.

As clearly pointed out, within the coherent coupling zone, the SLR resonances of the type-I and type-II gratings are virtually identical in terms of their profiles and wavelengths, as

can be seen in both the experimental and FDTD spectra (Figure 4). Note that the experimental spectra (Figure 4(b)) are broader than the FDTD spectra (Figure 4(a)), reflecting slight size variations in the fabricated gratings. As these figures show, the resonance profiles of the two types of gratings

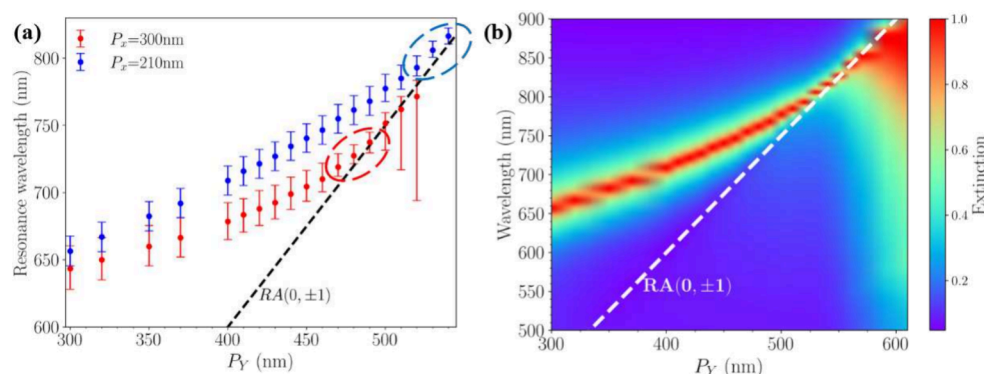


Figure 5. (a) Spectral position and fwhm of SLRs as function of P_y for type-I arrays in red and type-III arrays in blue. (b) Dispersion diagram of type-III gratings as a function of wavelength and P_y .

remain symmetrical as long as the zone of coherent coupling is not exceeded (a more energetic anomaly than the SLR). When $P_y = 500$ nm, the resonance maximum merges with the anomaly. Its width is greater than that at $P_y = 480$ and 490 nm, and the resonance profile is asymmetrical. Beyond this zone, the resonances become more asymmetrical, with a clear shoulder on the short-wavelength side. This reflects a reduced coupling with the anomaly $(0, \pm 1)$, which has become less energetic than the SLR.

As in Figure 4(a), we have superimposed in Figure 5 the positions of the maxima and half-widths of the calculated extinction spectra for type-I and type-III gratings (which have different filling factors). The closed, discontinuous contours in red and blue delimit the areas of coherent coupling of type-I and type-III arrays, respectively. The results for type-III arrays were derived from the corresponding dispersion diagram shown in Figure 4(b). Although the resonance profiles outside the coherence coupling zone are similar because of the identical size of the particles that make up the two types of grating, it is clear that the coherent coupling zones between the SLR resonances and the anomaly are not located in the same region. This behavior contrasts with that previously observed for type-I and type-II gratings, which share the same filling factor but differ in particle size, where coherent coupling with the anomaly occurs at exactly the same spectral location.

In this work, we confirmed by band calculation at the Γ point that SLRs are delocalized resonances dominated by their photonic character when spectrally located in the coherent coupling region, corresponding to the strong localization of electromagnetic energy in the dielectric space surrounding the metallic nanoparticles. Thus, we have shown that two gratings characterized by the same filling factor and lattice period transverse to the polarization direction of the exciting wave exhibit quasi-identical spectral energy density profiles located at the same wavelength. These results are supported by calculations and extinction measurements on the same gratings when illuminated at normal incidence by a wave polarized transversely to the grating period direction, which is kept constant for both gratings. This configuration is necessary to allow both gratings to support the same Rayleigh anomaly and thus bring the spectral zone of coherent coupling to the same location. As a result, the filling factor can be considered as an alternative parameter for localizing the coherent coupling zone. Controlling this parameter enables the design of new gratings composed of nanoparticles with different diameters and lattice periods depending on the direction of polarization, exhibiting the same SLR at the same wavelength. These different gratings

are characterized by the same filling factor and the same lattice period in the direction perpendicular to the polarization. Depending on the specific requirements of the desired applications, the gratings designed in this way will increase either the specific surface area of the nanoparticles or the area of strong electric field enhancement. These findings establish a simple geometric criterion—based on the filling factor—that enables the prediction and control of SLR wavelengths across diverse nanoparticle array geometries. It is noteworthy that our strategy has been applied in conditions where we are in normal incidence, with fixed particle height, and a homogeneous environment.

This principle offers a rational design pathway for tailoring plasmonic substrates in chemical sensing, photocatalysis, and surface-enhanced spectroscopy (SERS, SEIRA).

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Notes

The authors declare no competing financial interest.

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